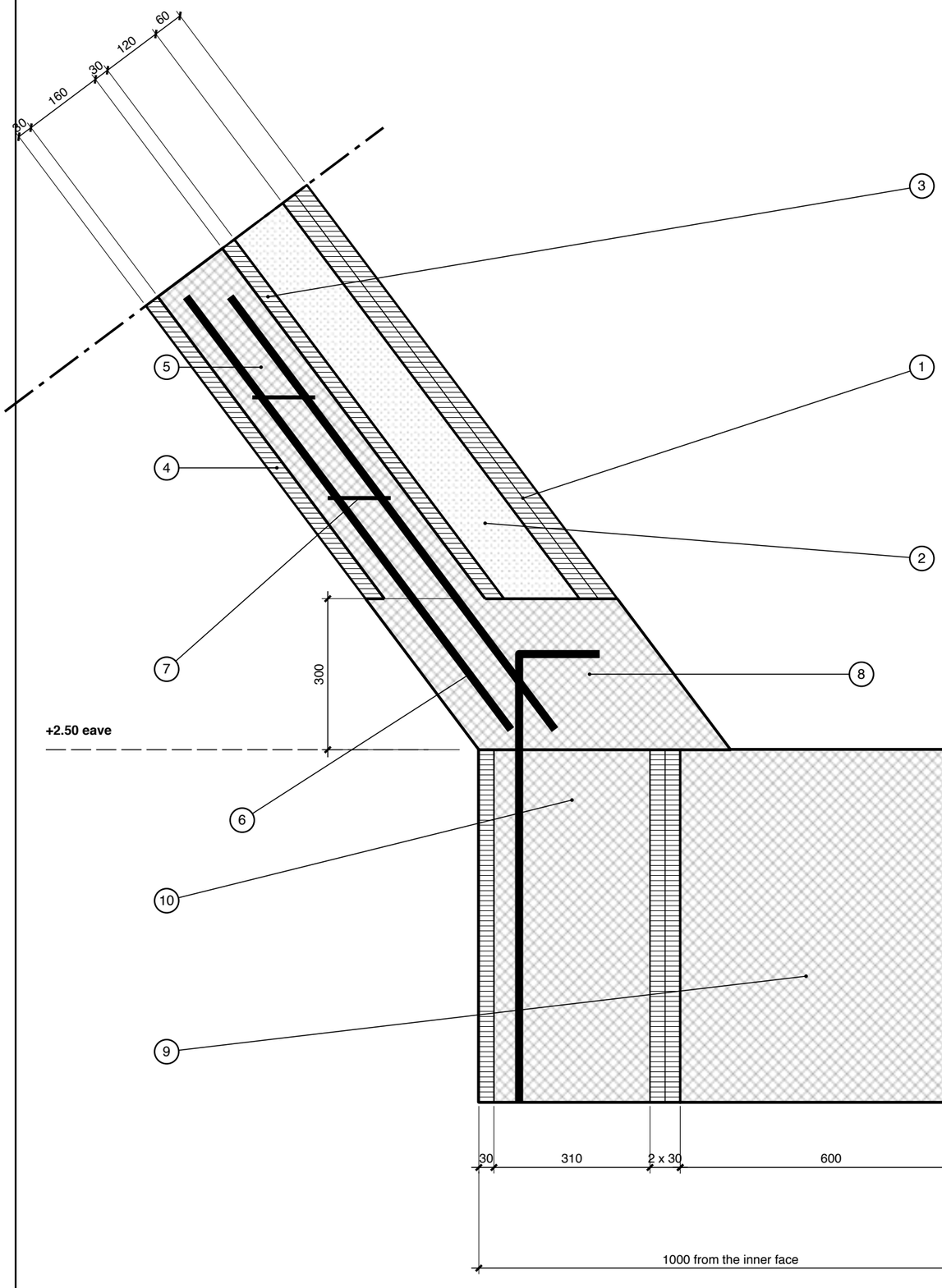
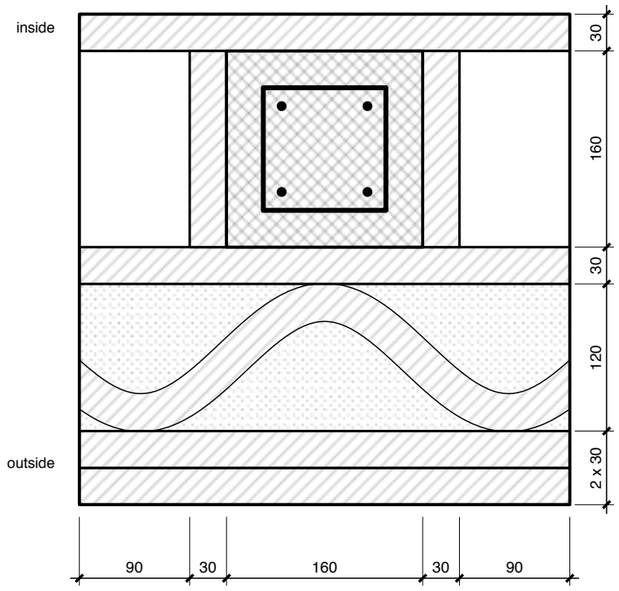
Detail D1: eave, roof to wall



Plan cut through a rib at one layer, 1:5



1. 2 outer beads 30, the outside of the printed piece: exposed concrete, no insulation or cladding outside (the insulation is in the lattice zone).
2. Lattice zone 120: a wavy printed bead 30 that touches the middle bead and the outer beads in turn (period about 300, assumed), insulation between its loops (dotted).
3. Middle bead 30: closes the rib zone on the outside. Inner and middle bead are the two structural skins of the roof, the ribs the webs between them (as in the Karamba model).
4. Inner bead 30, room side: continuous, it covers the ribs. The roof is 400 thick square to the slope from eave to crown.
5. Rib (cast column) 160 wide x 160 deep: fills the rib zone between the inner and the middle bead, inside two printed channel walls of 30 (see the plan cut), filled with poured concrete; section cut along the rib. It rises out of the solid base in the same pour, without a joint. Between the ribs the rib zone is an empty cavity.
6. Rib bars R, 4 Ø16 (2 in the cut), anchored in the solid base where they cross the hooks of the wall bars A (assumed; steel fibre in the concrete could replace them).
7. Stirrups Ø8 every 200 (assumed).
8. Solid base of the roof: the first 300 (vertical) above the eave is full cast concrete over the whole 400, no cavity or lattice, poured in one go with the rib above it; wall bars A hook into it. Replaces the ring beam.
9. Buttress 400 wide, 1000 from the inner face (600 beyond the wall): printed formwork (60 end cap of 2 beads shown) filled with cast concrete; top flush with the roof base at +2.50.
10. Wall 400 at the buttress: printed inner bead 30 and 2 outer beads 30, the 310 zone between them cast and one body with the buttress (no insulation here, the cut is through the buttress; between buttresses this zone is the wavy bead + insulation). Wall bars A Ø16 run in the cast zone from the footing up into the solid base and end in a 160 hook (assumed). The roof grid is A or B (see the roof-grid sheets); the detail is cut along a rib.

0 1 2 3 4 dm

Pyramid roof · 3D-printed concrete · GSAPP
Scale 1:10 at A3 · rib plan cut 1:5 · dimensions in mm
2026-10-14 · hand estimates, not engineered